Strategy to EA v2 — User Manual

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Turns one trading idea into three artifacts that agree with each other: a Technical Spec, a compilable MQL5 Expert Advisor for MetaTrader 5, and a Pine Script v6 `strategy()` for TradingView.

The framework enforces four guarantees:

1. **No contradictory options inside an aspect** — a field that a sibling option makes meaningless is greyed out and left out of every output.
2. **No contradictory options across aspects** — the market lens restricts which strategy families make sense, the family restricts the mechanic, and so on. Ruled-out options stay visible with the reason.
3. **Every setting you select reaches all three outputs** — the coverage panel proves it, field by field.
4. **The rule that fired is always shown** — when the tool adjusts something for you, the Report says which rule did it and why.

How the constraint engine works

Two mechanisms, both visible in the UI:

Field applicability. A field is either *in play* or *out of play*. Out-of-play fields are greyed with the reason, and they are never written into the spec or either script. `Stop distance (points)` only exists when the stop-loss mode is *Fixed points*.

Option availability. Inside a field, individual options can be ruled out by earlier choices. They stay on screen, struck through, with the reason. With the **Range** lens selected, *Breakout* and *Momentum* are ruled out because a range has no expansion to follow.

When a change makes your current selection illegal, the engine moves it to the nearest legal value and records the adjustment in the Report — it never silently rewrites your choice.

The ten aspects

#	Aspect	Question
1	**Target Market**	What instrument, timeframe, and session?
2	**Market Lens**	What environment must the market be in?
3	**Strategy / Edge**	What is the core idea, and what math powers it?
4	**Entry Logic**	Exactly when and at what price do we open?
5	**Filters**	Which setups are rejected?
6	**Exit Logic**	When do we take profit and cut loss?
7	**Risk-Reward & Size**	How much per trade, and at what reward ratio?
8	**Risk Management**	How do we protect the whole account?
9	**Data & Execution**	How do orders reach the broker reliably?
10	**Test & Validate**	How do we prove the strategy actually works?

1 · Target Market

What instrument, timeframe, and session?

Beginners over-scope. Pick ONE symbol, ONE timeframe, ONE session — lock it down first. The session hours are only emitted when you require a session window.

Setting	Options / range	In play	Lands in
Instrument / Symbol	free text (default **EURUSD**)	always in play	MQL5 logic · Pine logic
Timeframe	M1 · M5 · M15 · M30 · **H1** · H4 · D1	always in play	MQL5 logic · Pine logic

Require session window	**Session only** / 24h allowed	always in play	MQL5 logic · Pine logic
Session start (server hour)	0 – 23 (default **8**)	only when the session window is only used when “Require session window” is on	MQL5 logic · Pine logic
Session end (server hour)	0 – 23 (default **16**)	only when the session window is only used when “Require session window” is on	MQL5 logic · Pine logic

2 · Market Lens

What environment must the market be in?

The lens is a gate, not a label — it is compiled into the entry condition. It also restricts which strategy families make sense: you cannot fade a trend or follow a breakout inside a range.

Setting	Options / range	In play	Lands in
Market regime (lens)	**Trend** · Range · Volatility · Session bias	always in play	MQL5 logic · Pine logic
ADX minimum (trend gate)	10 – 40 (default **25**)	only when the ADX trend gate only applies to the Trend lens	MQL5 logic · Pine logic
ADX maximum (range gate)	5 – 35 (default **20**)	only when the ADX range gate only applies to the Range lens	MQL5 logic · Pine logic
Volatility floor (ATR % of price)	0.02 – 1.5 (default **0.3**)	only when the volatility floor applies to the Trend and Volatility lenses	MQL5 logic · Pine logic

3 · Strategy / Edge

What is the core idea, and what math powers it?

Pick ONE family and ONE mechanic. The mechanic selects a real algorithm — every mechanic below emits different code. Parameters are inputs so you can optimise later.

Setting	Options / range	In play	Lands in
Logic family	**Breakout** · Momentum · Mean Reversion · SMC / ICT	always in play	MQL5 logic · Pine logic
Mechanic	**Donchian channel** · Opening range (ORB)	always in play	MQL5 logic · Pine logic
Channel / band length	5 – 100 (default **55**)	only when used by the Donchian channel and Bollinger reversion mechanics	MQL5 logic · Pine logic
Fast MA length	3 – 60 (default **20**)	only when used by the EMA cross mechanic	MQL5 logic · Pine logic
Slow MA length	8 – 200 (default **50**)	only when used by the EMA cross mechanic	MQL5 logic · Pine logic
Opening-range bars	2 – 30 (default **6**)	only when used by the opening-range mechanic	MQL5 logic · Pine logic
MACD fast EMA	3 – 30 (default **12**)	only when used by the MACD mechanic	MQL5 logic · Pine logic
MACD slow EMA	8 – 60 (default **26**)	only when used by the MACD mechanic	MQL5 logic · Pine logic
MACD signal EMA	2 – 20 (default **9**)	only when used by the MACD mechanic	MQL5 logic · Pine logic
Bollinger deviation (σ)	1 – 4 (default **2**)	only when used by the Bollinger reversion	MQL5 logic · Pine logic

		mechanic	
RSI period	2 – 30 (default **14**)	only when used by the RSI reversion mechanic	MQL5 logic · Pine logic
RSI oversold level	5 – 45 (default **30**)	only when used by the RSI reversion mechanic	MQL5 logic · Pine logic
RSI overbought level	55 – 95 (default **70**)	only when used by the RSI reversion mechanic	MQL5 logic · Pine logic
Minimum gap size (× ATR)	0 – 2 (default **0.3**)	only when used by the fair-value-gap mechanic	MQL5 logic · Pine logic
Sweep lookback (bars)	5 – 80 (default **20**)	only when used by the liquidity-sweep mechanic	MQL5 logic · Pine logic
ATR period	5 – 40 (default **14**)	always in play	MQL5 logic · Pine logic

4 · Entry Logic

Exactly when and at what price do we open?

Everything here reads CLOSED bars — repainting is the #1 killer of backtests. “One bar later” is the only delay option, and it asks the condition to still hold.

Setting	Options / range	In play	Lands in
Direction	**Both** · Long only · Short only	always in play	MQL5 logic · Pine logic
Confirmation	**Signal bar** · One bar later	always in play	MQL5 logic · Pine logic
Order type	**Market** · Limit · Stop	always in play	MQL5 logic · Pine logic
Entry offset (× ATR)	0.1 – 5 (default **1**)	only when pending orders need an offset — a market	MQL5 logic · Pine logic

		order fills at the touch	
Slippage tolerance (points)	0 – 100 (default **10**)	always in play	MQL5 logic · Pine logic
Only trade with trend	**Only with trend** / Fade allowed	only when the Range lens already requires a non-trending market, so a trend-alignment filter would cancel every entry	MQL5 logic · Pine logic
Trend reference (SMA bars)	10 – 200 (default **50**)	only when the trend reference is only used by the trend-alignment filter	MQL5 logic · Pine logic

5 · Filters

Which setups are rejected?

Filters cut the trades that look good on a chart but bleed in reality. Every filter here is compiled into the entry condition, and the EA logs each rejection.

Setting	Options / range	In play	Lands in
Max spread (points, 0 = off)	0 – 100 (default **25**)	always in play	MQL5 logic · Pine documented
Max entries per day (0 = off)	0 – 20 (default **0**)	always in play	MQL5 logic · Pine logic
Close positions at session end	**Hold overnight** / Flat at close	only when there is no session boundary to close at when the session window is off	MQL5 logic · Pine logic

6 · Exit Logic

When do we take profit and cut loss?

Decide the stop first — it defines the risk. The target is then either a multiple of that risk or a fixed distance, never an independently-tuned number that disagrees with the stop.

Setting	Options / range	In play	Lands in
Stop-loss mode	**ATR multiple** · Fixed points · Swing structure	always in play	MQL5 logic · Pine logic
Stop distance (× ATR)	0.5 – 5 (default **2**)	only when the stop is currently measured in points or structure, not ATR	MQL5 logic · Pine logic
Stop distance (points)	5 – 500 (default **50**)	only when the stop is currently measured in ATR or structure, not points	MQL5 logic · Pine logic
Swing lookback (bars)	3 – 60 (default **20**)	only when the swing lookback only applies to the Swing structure stop	MQL5 logic · Pine logic
Take-profit mode	**Risk multiple (R)** · Fixed points	always in play	MQL5 logic · Pine logic
Target distance (points)	5 – 1000 (default **150**)	only when the target is currently a risk multiple, not a fixed distance	MQL5 logic · Pine logic
Trailing stop	Off · **ATR trail** · Points trail	always in play	MQL5 logic · Pine logic
Trail distance (× ATR)	0.5 – 4 (default **1.5**)	only when trailing is off or measured in points	MQL5 logic · Pine logic
		only when	

Trail distance (points)	5 – 300 (default **40**)	trailing is off or measured in ATR	MQL5 logic · Pine logic

7 · Risk-Reward & Size

How much per trade, and at what reward ratio?

Lots = (Equity × Risk%) ÷ (Stop distance in points × point value). The R target here is the only place the reward:risk ratio is set.

Setting	Options / range	In play	Lands in
Position size mode	**Risk-based %** · Fixed lots	always in play	MQL5 logic · Pine logic
Risk per trade (%)	0.1 – 5 (default **1**)	only when position sizing is set to fixed lots	MQL5 logic · Pine logic
Fixed lots	0.01 – 10 (default **0.1**)	only when position sizing is set to risk-based	MQL5 logic · Pine logic
Reward : risk target (R)	1 – 6 (default **3**)	only when the take-profit is a fixed distance, so no R multiple is used	MQL5 logic · Pine logic

8 · Risk Management

How do we protect the whole account?

These sit above position sizing and run every tick — a human cannot judge a kill switch in live time. Drawdown is measured peak-to-trough and latches when breached.

Setting	Options / range	In play	Lands in
Max daily loss (%)	1 – 20 (default **3**)	always in play	MQL5 logic · Pine logic
Max drawdown — halt (%)	5 – 50 (default **20**)	always in play	MQL5 logic · Pine logic
Max concurrent	1 – 5 (default	always in play	MQL5 logic ·

positions	**1**)		Pine logic

9 · Data & Execution

How do orders reach the broker reliably?

Over 90% of “EA does not trade” reports are execution or setup problems, not logic. Settings a Pine strategy cannot express are documented in its header rather than silently ignored.

Setting	Options / range	In play	Lands in
Magic number	free text (default **20240801**)	always in play	MQL5 logic · Pine documented
SL/TP placement	**Attached with order** · Separate modify	always in play	MQL5 logic · Pine logic
Send retries	0 – 10 (default **2**)	always in play	MQL5 logic · Pine documented
Retry delay (ms)	100 – 3000 (default **500**)	only when retry delay is meaningless with zero retries	MQL5 logic · Pine documented
On-chart dashboard	**No panel** / Show panel	always in play	MQL5 logic · Pine logic

10 · Test & Validate

How do we prove the strategy actually works?

Non-negotiables: no repainting, no look-ahead, closed bars only. A strategy with a 70% drawdown fails no matter the profit factor. These settings are written into the spec and both script headers — no script can set the tester for you.

Setting	Options / range	In play	Lands in
Backtest modelling	**Real ticks** · Every tick · 1-minute OHLC	always in play	MQL5 documented · Pine documented

Walk-forward split	**70 / 30** · 80 / 20 · None	always in play	MQL5 documented · Pine documented
Monte-Carlo runs	100 – 2000 (default **500**)	always in play	MQL5 documented · Pine documented
Minimum trades to trust	20 – 500 (default **100**)	always in play	MQL5 documented · Pine documented

What the lens does to the family

The market lens is not a label — it compiles into the regime gate, and it decides which strategy families are coherent:

Lens	Regime gate	Families allowed
Trend	ADX ≥ ADX minimum **and** ATR ≥ volatility floor	Breakout, Momentum, SMC / ICT
Range	ADX ≤ ADX maximum	Mean Reversion
Volatility	ATR ≥ volatility floor	Breakout, Momentum
Session bias	the session window itself	Breakout, Momentum, SMC / ICT

Each family offers two mechanics, and each mechanic emits different code:

Family	Mechanic	What it trades
Breakout	Donchian channel	Buy the break of the highest high over N bars, sell the break of the lowest low.
Breakout	Opening range (ORB)	Mark the high and low of the first N bars of the session, then trade the break of that range.
		Buy when the fast

Momentum	EMA cross	EMA crosses above the slow EMA, sell on the cross back below.
Momentum	MACD	Buy when the MACD line crosses above its signal line, sell on the cross below.
Mean Reversion	Bollinger reversion	Buy a close back inside the lower band, sell a close back inside the upper band.
Mean Reversion	RSI reversion	Buy when RSI crosses back up through the oversold level, sell when it crosses down through the overbought level.
SMC / ICT	Fair value gap (FVG)	Find a three-bar imbalance at least N × ATR wide, then trade the retest of the gap.
SMC / ICT	Liquidity sweep	Find a bar that takes out the N-bar extreme and closes back inside, then trade the reversal.

Where each setting lands

Pine Script v6 is generated as a real `strategy()`, so it is a peer of the EA rather than a drawing aid. Trailing stops become `trail_points` / `trail_offset`, the kill switches become `strategy.risk.max_intraday_loss` and `strategy.risk.max_drawdown`, pending orders become `strategy.entry(limit=…)` / `strategy.entry(stop=…)`, and position size is derived from `syminfo.pointvalue`.

Three settings have no Pine equivalent. Rather than drop them silently, they are written into the Pine script header and marked **documented** in the settings panel:

Setting	Why it cannot be code in Pine
Magic number	TradingView strategies have no

	order-tag concept
Max spread	Pine has no live spread feed; the value is modelled by `slippage`
Aspect 10 validation plan	No script can set a tester’s modelling mode or run count

Cautions

A caution is a combination that is legal but usually a mistake. It never blocks you — it states the consequence and offers a one-click fix where one exists.

Fires when	What it says
Daily loss limit ≥ drawdown halt	The daily limit can never be the one that fires first — the drawdown halt always trips earlier.
Risk % × max positions > 5%	Reports the total simultaneous exposure, e.g. 1% × 3 positions = 3% of the account at risk at once.
Risk per trade > 3%	Quantifies the five-loss streak cost, e.g. 4% per trade costs 18.5% of the account over five losses.
1-minute OHLC modelling on an M1/M5 chart	That model builds higher timeframes from M1 bars, so it cannot resolve the bars your stop sits inside.
No walk-forward split	Nothing out-of-sample validates the result, so an optimised backtest is unfalsifiable.
Stop-entry pending order + swing-structure stop	After a gap the swing stop can land on the wrong side of the pending entry level.
Entry offset > 2 × ATR	Most signals will never be filled — the live trade count will fall far short of the backtest.
M1 timeframe	M1 entries are dominated by spread and latency, so the backtest flatters the live result.
Max spread = 0	No spread limit is set, so entries can fill at poor prices on a wide-

	spread symbol.
Trail distance > stop distance	The trail starts wider than the initial stop, so it can only loosen the stop until price runs past it.
Session start = end	The window is treated as the full 24 hours.
Minimum trades > 300	Demanding that many trades usually forces a shorter timeframe or a looser filter, both of which degrade edge quality.
Max drawdown ≥ 40%	A halt that lets the account lose 40% before stopping; most funded-account rules cap total drawdown at 10%.

Using the generated code

MQL5 Expert Advisor

1. MetaEditor → **File ▸ New ▸ Expert Advisor**, name it, and paste the generated source over the template.
2. Press **F7** to compile.
3. Drag the EA onto an MT5 chart, tick **Allow Algo Trading**, and confirm the symbol and timeframe match the header comment.

The EA reads closed bars only, so a freshly attached EA waits for the next bar before it can act.

Pine Script v6

1. TradingView → **Pine Editor ▸ Open ▸ New strategy**.
2. Paste the generated source and press **Save**, then **Add to chart**.
3. Open the **Strategy Tester** tab to see the backtest.

Known limitations

- The EA and the Pine strategy are reference implementations of the same logic, not byte-equivalent.
- Aspect 10 is a plan, not code. It is written into the spec and both script headers and labelled *documented* in the settings panel.
- One symbol, one timeframe, one session per run. Run the tool again for another combination.
- The EA cancels its own pending orders when a new signal arrives

instead of expiring them on a timer.
- No walk-forward or Monte-Carlo engine is generated; those run outside the script.

Generated by Zo · Strategy → EA v2 workflow · 2026-10-19